

Popular Open Source Tools for Visualisation in AI/ML

Visualisation tools and techniques help to draw insights as well as find trends and patterns in AI/ML projects.. Open source tools are convenient and low cost solutions for converting raw AI/ML data into powerful visualisations. This article lists a few of these tools.



Data visualisations play a vital role in AI/ML projects, and help to get insights of data before and after the application of any machine learning model. Visualisation allows trends and patterns to be seen more effectively and easily, and is used for numerous purposes such as to analyse changeover time patterns, to determine the frequency of certain attributes, determine correlations among data, examine clusters of data, analyse the value and risks of underlining data, and schedule projects. There is a variety of charts that can be used in visualisation for different purposes. Line, Area, Bar, Histogram, Scatter, Bubble, Pie, Gauge, Map, Heat, Frame are popular visualisation charts in AI/ ML projects.

The best data visualisation tools include both OSV (open source visualisation) tools as well as proprietary licensed solutions. Open source visualisation tools are vital as they are free of cost and provide similar kinds of functionalities as proprietary

software. Table 1 describes the advantages of open source visualisation tools based on a number of parameters.

OSV tools for AI/ML

- **Hiplot:** This is a very powerful and lightweight tool for high dimensional data. It exists as several models – as a Web server in Jupyter Notebook and

as a command line interface (CLI). It represents information clearly by using features of parallel plots and other graphical utilities. It is vital for machine learning researchers as it evaluates the influence of their hyper parameters easily and effectively. This tool has various advantages in comparison with other tools. The main advantages are:

- **Simple to use:** It has three main variants: Web server, Jupyter Notebook and CLI, and is therefore suitable for many users. A simple syntax makes it easy to use for beginners.
- **Extendibility:** It provides default as well as custom parser facility. By default, the Web server of Hiplot parses CSV and JSON files. It is compatible with a number of AI libraries.
- **Interactive visualisation:** Due to its interactive visualisation feature, we can change the visualisation for different use cases.

Parameters	Benefit
Features/ Functionalities	Most OSV tools consist of numerous features/ functionalities and are applicable for all aspects of visualisation
Complex data handling	These tools handle complex data efficiently and are suitable for current needs of Big Data
API support	Most tools provide excellent API support
Ease of operation	OSV tools are command line or GUI based, and easy to operate
Development and updates	The best part about open source visualisation tools is that the development community is always making them better
Integration	Can be easily integrated with third party software and reporting dashboards
Pricing	Free of cost
Support	Lot of online tutorials and websites are available to ask questions from the community

Table 1: Benefits of open source visualisation tools